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SUMMARY

Aspiring Computer Science Engineer, seeking an internship position to apply and enhance my skills in AI, ML and VR. Passionate about developing innovative solutions and gaining practical experience in a dynamic environment.

PROJECTS

Virtual Reality Project Inspired by Picasso | UTC

July 2024

Compiègne, France

Overview:

Created a VR experience with a character inspired by Picasso's disfigured art style.

Key Achievements:

- Developed a custom VR character with asymmetric limb, modeled in Blender.
- Simulated a realistic limping effect using a cosine function.
- Applied inverse kinematics for precise hand and head tracking, enhancing immersion.
- Executed the project in Unity, demonstrating skills in 3D modeling, scripting, and real-time rendering.

Deepfake Detection using Meso4 Architecture | NUS

June 2024

Singapore

Overview:

Developed a deepfake detection system using the Meso4 architecture, inspired by the research paper "[A Compact Facial Video Forgery Detection Network](#)".

Key Achievements:

- Achieved 94.43% testing accuracy and 88.9% validation accuracy through pre-trained weights and hyperparameter tuning.
- Fine-tuned the model by unfreezing specific layers and using dropout layers to prevent overfitting.
- Utilized data augmentation techniques, including rescaling, shearing, zooming, and horizontal flipping, to enhance the diversity of the training data and improve model performance.

Interactive Virtual Reality Room

Overview:

Built an immersive virtual environment where users can freely move, interact with objects, and experience visually engaging scenes.

Key Achievements:

- Successfully implemented XR-enabled interactions (grabbing, throwing, wearing items) and snapping sockets for seamless object placement.
- Integrated both continuous movement and teleportation with fade transitions to minimize motion sickness and cater to different user preferences.
- Developed world-space canvases for settings and in-game menus, providing an intuitive and easily accessible interface in VR

Run/Walk Classifier

Overview:

Built a deep learning-based action classifier to distinguish between actions running and walking actions in images or videos.

Key Achievements:

- Integrated pre-trained VGG16 model, effectively capturing robust feature representations.
- Employed Keras Tuner (RandomSearch) to systematically identify optimal architecture parameters, improving validation accuracy to over 84%.
- Achieved a final test accuracy of 86%, demonstrating excellent predictive capabilities on unseen data.
- Implemented a pipeline for processing and classifying actions in video streams, enabling practical, real-time applications.

Arrhythmia Detection and Classification

Overview:

Performed a comparative analysis of cardiac arrhythmia detection and classification using machine learning models (Logistic Regression, Random Forest, Gradient Boosting, SVM, Decision Trees). Utilized Boruta for effective feature selection, enhancing model performance and interpretability.

Key Achievements:

- Achieved an accuracy of 87% using a Random Forest model for binary classification of arrhythmia.
- Improved classification performance with XGBoost, obtaining high sensitivity across multiple arrhythmia subtypes.
- Utilized the Boruta algorithm to identify critical predictors, thereby enhancing model reliability and interpretability.
- Compared various algorithms (e.g., logistic regression, SVM, decision trees, KNN, neural networks) and used cross-validation to fine-tune models for optimal performance.
- Delivered a solution with potential to aid healthcare professionals by reducing misdiagnoses and improving early detection of life-threatening arrhythmia.

TECHNICAL SKILLS

- Programming Languages: Python, C++, C#, Java, HTML, CSS, Javascript, React
- Frameworks and Libraries: TensorFlow, Keras, Numpy, Matplotlib
- Tools: Git, Jupyter Notebook, Unity, XR Interaction Toolkit, OpenXR
- Other Skills: Hyperparameter Tuning, Inverse Kinematics, Project Management

EDUCATION

Vellore Institute of Technology

Aug 2023 - Aug 2027

B.Tech | Computer Science Engineering. | GPA: 8.66

EXTRACURRICULAR ACTIVITIES

Senior Core Member, IEEE RAS Club

Dec 2024

- Assisted in organizing an Ideathon for 100 participants and a Hackathon for 250 participants, fostering innovation and problem-solving among students.
- Designed and conducted a hands-on robotics event for freshers, where participants built a line-following robot by solving technical puzzles and challenges.
- Conducted multiple workshops for freshers on AI/ML, Embedded Systems, and Web Development, providing hands-on learning experiences and mentorship.